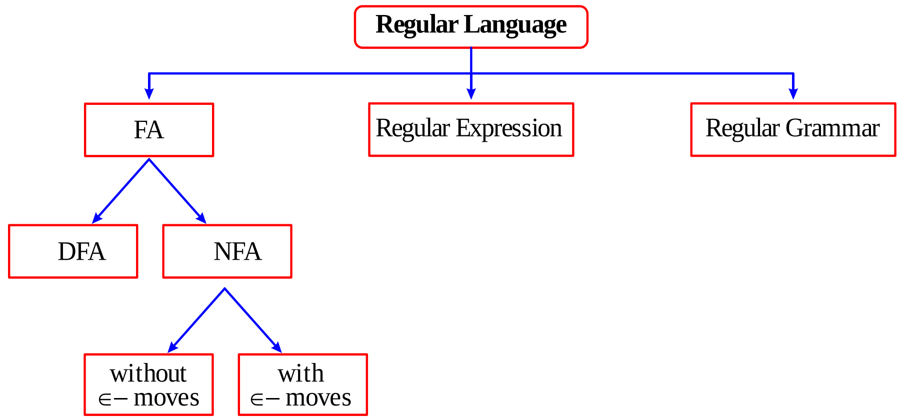
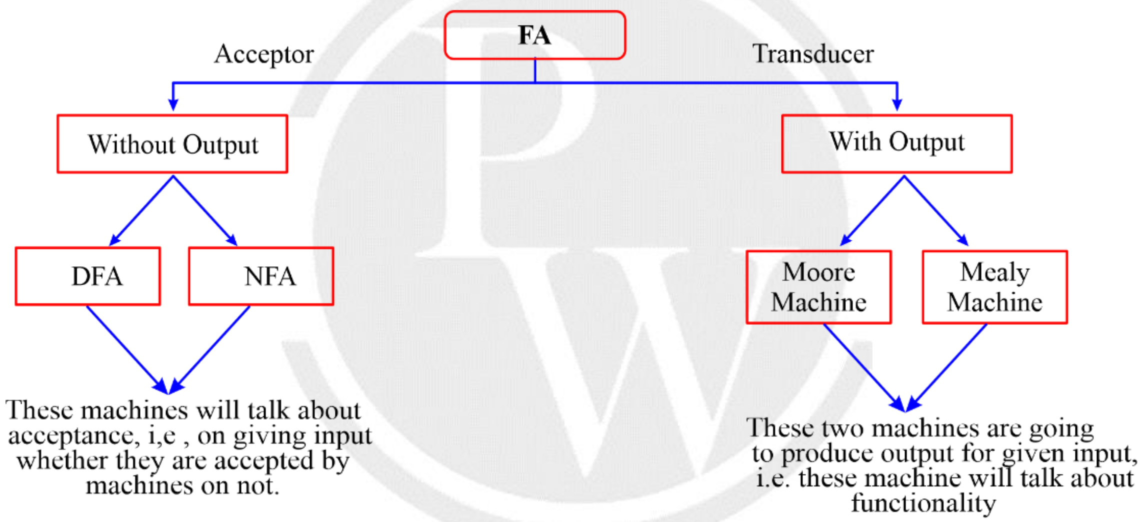
2.6 Regular language Representation



2.7 FA Classification

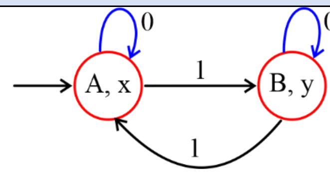


2.8 Moore machine and mealy machine

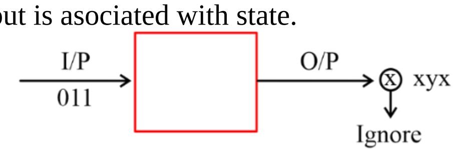
Moore machine	Mealy machine
Transition Function $\delta : Q \times \Sigma \to Q$ Output Function $\lambda : Q \to \Delta$ Outpput is associated with every state.	$\delta : Q \times \Sigma \to Q$ $\lambda : Q \times \Sigma \to \Delta$ Output is associated with transition.

2.8.1 Example for Moore machine and Mealy machine

Moore



$Q = \{A, B\}$
$\Sigma = \{0, 1\}$
$\Delta = \{x, y\}$
Outpput is asociated with state.

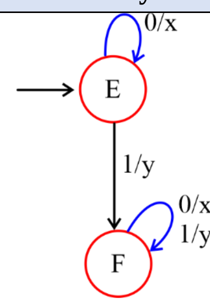


$$A \xrightarrow{0} A \xrightarrow{1} B \xrightarrow{1} A$$
$$x \quad x \quad y \quad x$$

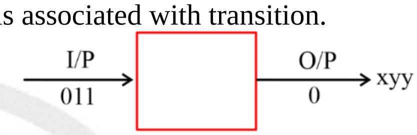
This is not expected.

- Extra one output is produced other than desired output.
- So, we can ignore this extra output as this is the machine property.

Mealy



$Q = \{A, B\}$
$\Sigma = \{0, 1\}$
$\Delta = \{x, y\}$
Output is associated with transition.



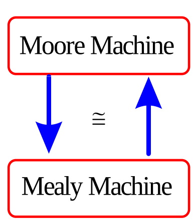
$$E \xrightarrow{0} E \xrightarrow{1} F \xrightarrow{1} F$$
$$x \quad y \quad y$$

- No extra output is produced.
- For 3 length input, we are getting the 3 length output.

2.9 Difference between Moore machine and Mealy machine

		Moore machine	Mealy machine
1.	δ	$Q \times \Sigma \to Q$	$Q \times \Sigma \to Q$
2.	λ	$Q \to \Delta$	$Q \times \Sigma \to \Delta$
3.	Length of O/p	If n length I/P (assume 1 length O/P symbol is taken at each state) then O/P length is (n+1).	If n length input (assume 1 length O/P symbol is taken at each transition) is given then O/P length is n.
4.	By default	DFA no final state.	DFA no final state.

2.10 Constuction of FA with output

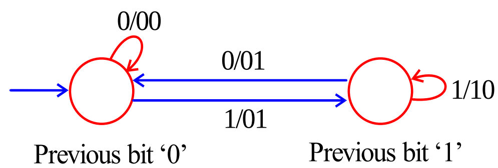


> **Note :**
>
> For every problem, if moore machine exists the we can also construct equivalent mealy machine.

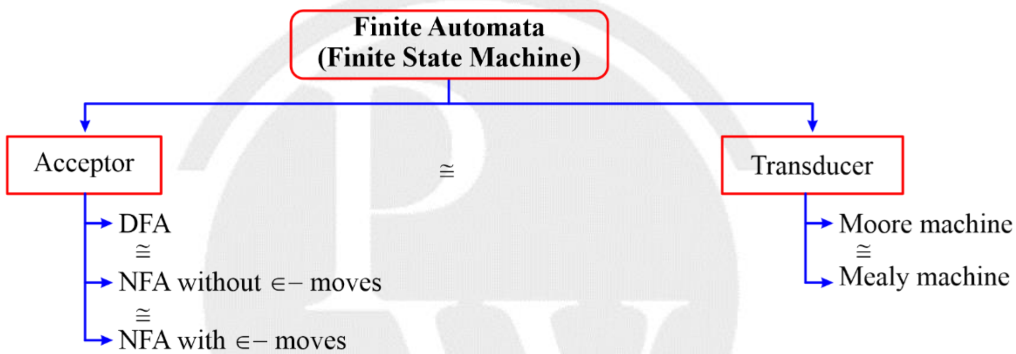
Example:

Sum of present bit and previous bit.

Mealy Machine



2.11 Classification of Finite State Machine (FSM)



2.12 Regular Expression

Definition:

- Regular expression represents a regular language.
- It describes a regular set.
- L (regular expression) is regular set.
- It is a kind of declarative way to represents a regular language.
- Regular expression generates a regular set.
- It uses 4 operators to represent a regular language.